

# Complex Analysis — Section 4.4

## Power Series Functions (Howell & Mathews, online)

# Section Overview

- Study of power series and their domains
- Classification of convergence sets
- Radius of convergence

# Definition: Power Series Function

A function defined by a series

$$f(z) = \sum_{n=0}^{\infty} c_n (z - \alpha)^n$$

is a power series function on its domain of convergence.

## Theorem 4.4.1: Convergence Sets

Suppose  $f(z) = \sum_{n=0}^{\infty} c_n(z - \alpha)^n$ . Then the set of  $z$  where the series converges is:

- 1 The single point  $z = \alpha$ ,
- 2 An open disk  $D_\rho(\alpha)$  with possible boundary points,
- 3 The entire complex plane.

# Radius of Convergence

Let  $\rho$  be the radius of convergence. Then we can find  $\rho$  by:

- Cauchy's root test:  $\rho = \frac{1}{\limsup |c_n|^{1/n}},$
- Ratio test (when limit exists):  $\rho = \frac{1}{\lim |\frac{c_{n+1}}{c_n}|}.$

## Examples (Quick Sketches)

- $\sum (\frac{n+2}{3n+1})^n (z-4)^n$  has radius  $\rho = 3$ .

# What This Means

Power series behave like polynomials locally but have well-defined maximal disks of convergence.